$\pi_3(S^2) = \mathbb{Z}$ BY A POSTNIKOV UNIVERSAL PROPERTY: INFORMAL PROOF, STATUS OF THE UNIT FORMALISATION

Status. This note answers `doc/pi3-s2-postnikov-handoff.txt`. The checked code is `sag/pi3-s2-postnikov` (`./segc sag/pi3-s2-postnikov all`: 0 failed; a fresh process from the directory: 2877 declarations, 0 failed, 1.1 s). *Update (28 Oct 2026):* $\pi_3(S^2) \cong \mathbb{Z}$ *is now proved from the sphere axiom alone,* `Presentation.Final.pi3-S2` *(71 modules, 3135 declarations, 0 failed), through the mapping theorem for targets* Fib$k$ *(*`doc/mapping-fib.pdf`*) and a Postnikov presentation of* $\|S^2\|_3$ *(*`doc/postnikov-presentation.pdf`*); the text below records the state before.* **The theorem $\pi_3(S^2) = \mathbb{Z}$ is not proved unconditionally.** What is checked is:

- (A) the sphere axiom, its pointed universal property $\mathrm{Map}_*(S^2, X) \simeq \Omega^2 X$ with naturality; pointed homotopy classes $[X, Y]_*$ as *sets*, with composition and the category laws; π_3 as a group, a functor on classes;
- (B) the universal-property argument, *conditional* on two explicit interfaces (a 3-truncation of S^2; the representing property of P), concluding $\pi_3(S^2) \cong \pi_3(P)$ as groups;
- (C, objects) for every Postnikov triple (A, B, k), the realization Fib$k$ is a simply connected 3-type and $\pi_3(\mathrm{Fib}k) \cong B$ as groups, unconditionally;
- (A′) finite truncations $\|X\|_m$ from univalence and replacement, and an inhabitant of the interface `Trunc3` for S^2, with the proof that S^2 is simply connected (`doc/finite-truncation.pdf`; `SphereTruncation.SphereTrunc.trunc3`);
- (H) the cohomology input, unconditionally: the cup square κ (`CupSquare.kappa`, `doc/cup-square.pdf`), (H3) (`H3Proof.h3`, `doc/circle-torsors.pdf`) and (H4) for this κ, every abelian B (`H4Proof.h4`, `doc/h4.pdf`);
- (E, conditional) for any $\kappa : K(\mathbb{Z}, 2) \to_* K(\mathbb{Z}, 4)$ and the mapping theorem for $P = \mathrm{Fib}\kappa$, $\pi_3(S^2) \cong \mathbb{Z}$ as groups (`Pi3Sphere.Conditional.pi3-S2-T`; the 3-truncation is the constructed one).

Open: the mapping theorem for the source P. The cohomology input (the construction of κ with (H4), and (H3); stated as types in `KZ2Cohomology`, not assumed) was open when this note was first written. *Update (28 Oct 2026):* (H3) is proved, `H3Proof.h3 : KZ2Cohomology.H3`, from the circle property of $K(\mathbb{Z}, 1)$ (`CircleProperty`; informal proof `doc/circle-torsors.pdf`); κ is constructed as the cup square (`CupSquare.kappa`, `doc/cup-square.pdf`) and (H4) is proved for it, `H4Proof.h4 : KZ2Cohomology.H4 CupSquare.kappa` (`doc/h4.pdf`). Sections 6 and 8 give the precise obligations. No Agda or other external source was inspected during this work; the references of the handoff were not consulted, and nothing here depends on them.

The formal files were written in the order of the milestones, before this note; the note is the informal proof that the files now follow, declaration by declaration (§7), in the sense of `seg/WRITING.md`.

1. Conventions

Unit's indexing (`HLevels`, `Connected`): `isTrunc n X` is "X is $(n-2)$-truncated", `isConn n X` is "X is $(n-2)$-connected". So a *3-type* is `isTrunc 5` and *simply connected* is `isConn 3`. For Eilenberg–MacLane spaces we use `EMMaps.KK[A].K m`, which is $K(A, m)$ with $K(A, 0) = A$; the offset of `EM.EMSpace.K` ($K\,n = K(A, n+1)$) does not appear in the new files. ΩX is $(x_0 = x_0, \mathsf{refl})$; $\Omega^2 X$ is `O2 X x0`. $\pi_3(X)$ is $\pi_1(\Omega^2 X)$ (`PiOne`: loops modulo mere equality, composition of the outer loop). $[X, Y]_*$ is the set truncation of $\mathrm{Map}_*(X, Y)$, i.e. its quotient by mere equality of *pointed* maps (`Quotient`, from replacement).

Date: 28 October 2026.

2. The sphere (milestone A)

Definition 1 (the authorised axiom, `SphereAxiom.Sphere`)**.** A type S^2 with base and $\mathsf{surf} : \Omega^2(S^2, \mathsf{base})$ such that for every X in the same universe the canonical map

$$\mathrm{ev}_2 : (S^2 \to X) \to \textstyle\sum_{x:X} \Omega^2(X, x), \qquad f \mapsto (f\,\mathsf{base},\ \mathsf{ap}(\mathsf{ap}f)\,\mathsf{surf})$$

is an equivalence. ($\mathsf{ap}f\,\mathsf{refl}$ is refl by the computation rule of J, so the second component is a 2-loop at $f\,\mathsf{base}$.)

This is recursion with its uniqueness (the non-dependent universal property of the HIT), stated through the one canonical map; it mentions no homotopy group. The axiom is the parameter telescope of the segment `Sphere`, not a parameter of `Theory`.

Lemma 2 (`evPt`, `pointed-up`, `evPt-nat`)**.** *Let* $\mathrm{ev}_*(F) = (\Omega^2 F)(\mathsf{surf})$ *for* $F : S^2 \to_* X$*. Then* $\mathrm{ev}_* : \mathrm{Map}_*(S^2, X) \to \Omega^2 X$ *is an equivalence, and* $\mathrm{ev}_*(G \circ F) = (\Omega^2 G)(\mathrm{ev}_* F)$*.*

Proof. Step 1 (`Refl.eq`). For $F = (f, \mathsf{refl})$ the looping $J = \Omega F$ and $K = (\mathsf{ap}f, \mathsf{refl})$ are homotopic by $H\,q : J\,q = \mathsf{refl} \cdot \mathsf{ap}f\,q = \mathsf{ap}f\,q$ (`loop-ap`, `trans-refl-l`). By `LoopConj.loop-homotopy`, $(\Omega K)\,\mathsf{surf} = c^{-1}\,((\Omega J)\,\mathsf{surf})\,c$ with $c = J_0^{-1} \cdot H(\mathsf{refl}) \cdot K_0$, and $c = \mathsf{refl}$ by computation (`c-refl`, stated by `refl`). So $\mathrm{ev}_* F = \mathsf{ap}(\mathsf{ap}f)\,\mathsf{surf}$. *Step 2* (`evPt-subst`). By J on the pointing p, $\mathrm{ev}_*(f, p) = p_*(\mathsf{ap}(\mathsf{ap}f)\,\mathsf{surf})$. *Step 3* (`pointed-up`). $\mathrm{Map}_*(S^2, X) = \sum_f (f\,\mathsf{base} = x_0)$ is, by Σ over the equivalence ev_2 of bases (`sigma-base`: the first component of $\mathrm{ev}_2 f$ is $f\,\mathsf{base}$ by definition), equivalent to $\sum_{(x,\alpha)}(x = x_0)$, which the transport $((x, \alpha), p) \mapsto p_*\alpha$ identifies with $\Omega^2 X$ (`Contract`: singleton contraction). The composite is ev_* up to Step 2. *Step 4* (`evPt-nat`). $\Omega(G \circ F) = \Omega G \circ \Omega F$ twice (`LoopCalc.loop-comp`). □

3. Homotopy classes and π_3 (milestone A)

Lemma 3 (`HomotopyClasses`)**.** $[X, Y]_*$ *is a set; composition descends in both arguments (`compC`, with `comp-beta`:* $[F];[G] = [F;G]$*); it is associative and unital (`comp-assoc`, `comp-idl`, `comp-idr`, from the laws for pointed maps). If precomposition with* F *is an equivalence of mapping types, precomposition with* $[F]$ *is an equivalence of sets of classes (`pre-equiv`, via `classes-equiv`).*

Equality of classes is *mere* equality of pointed maps (`STr.effective`); nothing is said about the type of homotopies.

Lemma 4 (`HomotopyGroups`)**.** π_1 *is a functor on pointed maps (`ind-comp`, `ind-id`);* $\Omega^2(F;G) = \Omega^2 F;\Omega^2 G$ *and* $\Omega^2\mathrm{id} = \mathrm{id}$ *(`loop2-comp`, `loop2-id`); hence* π_3 *is a functor on pointed maps. Homomorphisms form a set (`ghom-set`), so* $F \mapsto \pi_3 F$ *descends to classes (`pi3C`), functorially (`pi3C-comp`, `pi3C-id`).*

In particular π_3 is applied to a class without choosing a representative; this is what the two-map argument below needs.

4. The universal-property argument (milestone B, conditional)

Definition 5 (`SphereUniversal.Interfaces`)**.** *A 3-truncation of* S^2 (`Trunc3`): a pointed $\eta : S^2 \to_* T$ with T a simply connected 3-type, such that precomposition with η is an equivalence $\mathrm{Map}_*(T, Y) \simeq \mathrm{Map}_*(S^2, Y)$ for every 3-type Y, and $\pi_3\eta$ is bijective. *A representing object* (`Rep`): a simply connected 3-type P with a class $s \in [S^2, P]_*$ such that, for every simply connected 3-type X, $s^* : [P, X]_* \to [S^2, X]_*$ is a bijection of sets.

By Lemma 2, $[S^2, X]_* = \|\Omega^2 X\|_0 = \pi_2 X$ (as sets), so `Rep` is "$[P, X]_* \cong \pi_2 X$ naturally in X", the normalisation sending $[F]$ to $\pi_2 F$ applied to the element of $\pi_2 P$ given by s. The marked object is (P, s); P alone is not an initial object.

Theorem 6 (`Assemble.iso-S2-P`)**.** *Given* ***Trunc3*** *and* ***Rep****,* $\pi_3(S^2) \cong \pi_3(P)$ *as groups.*

Proof. Let $e = [\eta]$. Precomposition with e is a bijection on classes into P and into T (`etaP`, `etaT`: `pre-equiv` and `Trunc3.up`); s^* is a bijection on classes into P and T (`sP`, `sT`: `Rep.rep`). Let $u \in [T, P]_*$ with $e; u = s$ and $t \in [P, T]_*$ with $s; t = e$ (the inverses at s, resp. e). Then

$$s; (t; u) = (s; t); u = e; u = s = s; \mathrm{id}_P, \qquad e; (u; t) = (e; u); t = s; t = e = e; \mathrm{id}_T,$$

so $t; u = \mathrm{id}_P$ and $u; t = \mathrm{id}_T$ by injectivity (`tu`, `ut`). Applying π_3 on classes, $\pi_3 u$ and $\pi_3 t$ are inverse homomorphisms (`iso-T-P`). With $\pi_3\eta$, bijective and hence an isomorphism (`gi-of-equiv`: the inverse of a bijective homomorphism is multiplicative), we get $\pi_3 S^2 \cong \pi_3 T \cong \pi_3 P$. □

Only $X = T$ and $X = P$ are used in `Rep.rep`; the general statement is kept because it is what the mapping theorem gives.

5. Postnikov triples (milestone C, objects)

A triple is (A, B, k) with A, B abelian groups and $k : K(A, 2) \to_* K(B, 4)$; its realization is the pointed fibre $\mathrm{Fib}k = \sum_x (k\, x = \mathtt{pt})$.

Proposition 7 (`PostnikovRealization.Triple`)**.** Fib$k$ *is a 3-type (****trunc5****) and simply connected (****conn3****), and* $\pi_3(\mathrm{Fib}k) \cong B$ *as groups (****pi3-iso****).*

Proof. Levels: a Σ of a 2-type and of identity types of a 4-type. Connectedness: Fib$k$ is 0-connected since every loop of $K(B, 4)$ is merely `refl` (`Fibres.FibConn`, `surj-triv`); its loop space is $\mathrm{Fib}(\Omega k)$ (`omega-fib`), 0-connected for the same reason one level down, so Fib$k$ is 1-connected (`conn-from-point`).

π_3: a chain of isomorphisms, each induced by a pointed map whose looping is an equivalence (`LoopIso.LoopEquivIso`: the inverse on π_1 descends from the inverse on loops; no choice):

$$\begin{aligned}
\pi_1\Omega^2\mathrm{Fib}k &\cong \pi_1\Omega\mathrm{Fib}(\Omega k) && (\Omega \text{ of } \mathtt{omega\text{-}fib};\ \mathtt{s1})\\
&\cong \pi_1\mathrm{Fib}(\Omega^2 k) && (\mathtt{omega\text{-}fib};\ \mathtt{s2})\\
&\cong \pi_1\mathrm{Fib}(\mathrm{pr} : \mathrm{Fib}(\Omega^2 k) \to \Omega^2 K(A,2)) \cong \pi_1\Omega^3 K(B,4) && (\mathtt{ProjIso},\ \mathtt{puppe1};\ \mathtt{s3})\\
&\cong \pi_1 K(B,1) && (\text{loops of } \mathtt{KK.e};\ \mathtt{s4})\\
&\cong B && (\text{the band, a multiplicative bijection};\ \mathtt{s5}).
\end{aligned}$$

For `s3`: $\Omega^2 K(A, 2)$ is a set, so the projection $\mathrm{Fib}(\mathrm{pr}) \to \mathrm{Fib}(\Omega^2 k)$ is a subtype inclusion (its fibres are identity types of a set), whose `ap` is an equivalence (`ap-fst-equiv`); and $\Omega^3 K(B, 4) \simeq \mathrm{Fib}(\mathrm{pr})$ is the Puppe equivalence. This is the boundary $\pi_4 K(B, 4) \to \pi_3\mathrm{Fib}k$ of the fibre sequence, read through iterated looping of fibres, preserving the group operation at each step. □

Corollary 8 (`Pi3Sphere.Conditional.Final.pi3-S2`)**.** *For the sphere axiom, any* $\kappa : K(\mathbb{Z}, 2) \to_* K(\mathbb{Z}, 4)$*, a* ***Trunc3*** *for* S^2 *and a* ***MapThm*** *for* $P = \mathrm{Fib}\kappa$ *(a class* $s \in [S^2, P]_*$ *with* s^* *bijective onto* $[S^2, X]_*$ *for all simply connected 3-types* X*):* $\pi_3(S^2) \cong \mathbb{Z}$ *(****Integers.intAb****) as groups. (Theorem 6 and Proposition 7.)*

This is the final signature in conditional form: its parameters `kappa`, `TR`, `MT` are the open obligations below. They are explicit parameters of a segment, not Theory assumptions.

6. Open obligations

Obligation 9 (finite truncation; feeds `Trunc3`. **DONE**, see `doc/finite-truncation.pdf`)**.** Construct $\|S^2\|_3$ from univalence and replacement, with: the 3-truncation bound, the unit, the universal property for maps into 3-types (pointed form), $\Omega^n\|X\|_n \simeq \|\Omega^n X\|_0$ for $n = 3$ (so $\pi_3\eta$ is bijective), simple connectivity (which needs that S^2 is 1-connected). A route: Rijke's construction

of $\|A\|_{n+1}$ as the image of $a \mapsto \|a = -\|_n : A \to (A \to U_{\leq n})$, by induction from set truncation (`Quotient`); the identity types of the image are $\|a = b\|_n$ by univalence and the Yoneda lemma for n-truncated families, and the image is small by replacement (`Completion`). The universal property for maps into $(n+1)$-types then has to be proved; this is the part of the route still to be written out in detail. No truncation axiom has been added. *Update:* carried out in `ConnectedMaps`, `Truncations`, `TruncationProps`, `SphereTruncation`. The universal property is obtained from the connectivity of the unit by an elimination lemma for connected maps, proved with the inductive definition of connectedness, so no join and no connectivity theorem of the paper is needed.

Obligation 10 (the cohomology input; `KZ2Cohomology`. **DONE**, see `doc/cup-square.pdf`, `doc/h4.pdf`)**.** Construct κ and prove, for every abelian group B,

(H4) $\Phi_B : \mathrm{Hom}(\mathbb{Z}, B) \to [K(\mathbb{Z},2), K(B,4)]_*$, $g \mapsto [\kappa; K(g,4)]$ is a bijection, (H3) $[K(\mathbb{Z},2), K(B,3)]_*$ is cor

Both are stated as types (`H4 kappa`, `H3`); neither is assumed. *Update:* (H3) is now proved (`H3Proof.h3`) along the first reduction below, with the circle property of the $\mathbb{Z}$-torsors proved in `CircleProperty` (`doc/circle-torsors.pdf`). (H4) is proved for the cup square κ (`H4Proof.h4`, `doc/h4.pdf`): the second reduction below is Proposition 5 there (`H4Classes`); the remaining content is obtained without pushouts and without the quadratic-map route, as $\Phi_B = \mathrm{cls} \circ \Delta' \circ (g \mapsto c_2; K(g,4)_*)$, a composite of bijections: every class is the diagonal of a unique bipointed map (Theorem A, `H4Diagonal`) and $g \mapsto c_2; K(g,4)_*$ is a bijection onto the bipointed maps (Theorem B, `H4Bilinear`), for every abelian B, $B = \mathbb{Z}/2$ included. The paragraphs below record the state before that proof. Informal reductions:

- (H3). Looping is a map $\mathrm{Map}_*(K(\mathbb{Z},2), K(B,3)) \to \mathrm{Map}_*(K(\mathbb{Z},1), K(B,2))$ whose fibres (the deloopings) are propositions, by DW1 Cor. 10 ($K(\mathbb{Z},2)$ is 1-connected, $K(B,3)$ is 3-truncated: $2 \cdot 1 + m + 2 = 3$, $m = -1$), so it is an embedding. If $K(\mathbb{Z},1)$ (the $\mathbb{Z}$-torsors) has the universal property of the circle, $\mathrm{Map}_*(K(\mathbb{Z},1), K(B,2)) \simeq \Omega K(B,2) \simeq K(B,1)$ is connected. A subtype of a connected type containing the base point has one component, so $[K(\mathbb{Z},2), K(B,3)]_*$ is a point. The missing lemma is the circle property of $\mathbb{Z}$-torsors (Bezem–Buchholtz–Grayson–Shulman); it is not in the library.
- (H4). The same looping, now with fibres of level $m = 0$ (sets), maps $\mathrm{Map}_*(K(\mathbb{Z},2), K(B,4))$ to $\mathrm{Map}_*(K(\mathbb{Z},1), K(B,3)) \simeq K(B,2)$, which is simply connected; so $[K(\mathbb{Z},2), K(B,4)]_*$ is the set of deloopings of the zero map $K(\mathbb{Z},1) \to_* K(B,3)$. That this set is B, with the coefficient changes, is the real content (classically the Hopf construction, or $H^4(\mathbb{CP}^\infty)$). I do not have a synthetic proof of it from the authorised foundations: the cellular route through $\mathbb{CP}^\infty$ needs pushouts, which are not authorised. This is the main open mathematical obligation. The quadratic-map route of the handoff ($q(-a) = q(a)$, bilinear cross-effect; $\Gamma(\mathbb{Z}) = \mathbb{Z}$, generator with $q(1)$ a generator, not $2q(1)$) would equally need a comparison theorem with pointed maps $K(A,2) \to K(B,4)$, which is not available.

Obligation 11 (the mapping theorem for the source $P = \mathrm{Fib}\kappa$; feeds `MapThm`)**.** Informal proof, given (H3), (H4) and the obligation on truncations, for a target $X = \mathrm{Fib}k$ with $k : K(A,2) \to_* K(B,4)$:

(1) $\mathrm{Map}_*(P, \mathrm{Fib}k) \simeq \sum_{G:\mathrm{Map}_*(P,K(A,2))} (G; k =_* \mathrm{const})$ (universal property of the pointed fibre).
(2) Precomposition with $\mathrm{pr} : P \to K(\mathbb{Z},2)$ is an equivalence $\mathrm{Map}_*(K(\mathbb{Z},2), K(A,2)) \simeq \mathrm{Map}_*(P, K(A,2))$: the fibres of pr are merely $K(\mathbb{Z},3)$ (2-connected) and $K(A,2)$ is a 2-type (Lemma 4 in the pointed form, fibrewise). And $\mathrm{Map}_*(K(\mathbb{Z},2), K(A,2)) \simeq \mathrm{Hom}(\mathbb{Z}, A) \simeq A$ is a set (unique deloopings, DW1 Cor. 11; evaluation at 1).

(3) Hence $[P, \mathrm{Fib}k]_* \cong \sum_{a:A} \pi_0 \,\mathrm{Null}(h_a)$ with $h_a = \mathrm{pr}; K(a,2); k$. By (H4), $K(a,2); k = \kappa; K(g,4)$ merely, for a unique g, so h_a is merely $\mathrm{pr}; \kappa; K(g,4)$, which is null by the fibre sequence. A path space in $\mathrm{Map}_*(P, K(B,4))$ that is inhabited has π_0 a torsor under $[P, K(B,3)]_*$ (translation by the additive structure of $K(B,4)$, or equivalently path composition); so it is a point once $[P, K(B,3)]_* = 0$.

(4) $[P, K(B,3)]_* = 0$: from the low-degree exact sequence of the fibration $K(\mathbb{Z},3) \to P \to K(\mathbb{Z},2)$, $0 = H^3(K(\mathbb{Z},2); B) \to H^3(P; B) \to H^3(K(\mathbb{Z},3); B) \xrightarrow{\tau} H^4(K(\mathbb{Z},2); B)$, with $\tau = \Phi_B$ injective by (H4). The exactness (a synthetic Serre sequence in degrees ≤ 4, base 1-connected, fibre 2-connected) is itself to be proved. *Update (28 Oct 2026):* **proved**, without a Serre sequence, for $P = \mathrm{Fib}\kappa$ and any κ with (H4): `FibreH3.fib-h3` (`doc/fibre-h3.pdf`, refereed). The restriction of F to the fibre deloops to G; F gives an explicit null-homotopy of $\kappa; G$ (so the connecting map is Φ_B by construction); $G = K(g,4)$ and (H4) give $G = 0$; then F factors through $K(\mathbb{Z},2)$ and (H3) kills it.

(5) Naturality in X and the reduction of an arbitrary X in C to a presentation $\mathrm{Fib}k$: the statement "s^* is bijective at X" is a proposition, so a mere presentation suffices (no global choice). The presentation needs $X \to \|X\|_2 \simeq K(\pi_2 X, 2)$, the fibre a 2-connected 3-type $\simeq K(\pi_3 X, 3)$, and the classification of the resulting fibration over a simply connected base by a map to $K(\pi_3 X, 4)$. For Theorem 6 only $X = P$ (a presentation by definition) and $X = T$ are needed.

None of this is formalised. The handoff's triple-morphism formulation (f, g, H with H a pointed homotopy, the H^3-ambiguity removed by (H3) on components, not on homotopies) is equivalent for this source; the reduction above was preferred because it avoids building the general morphism theory.

7. Correspondence with the formal declarations

Lemma 2	`SphereAxiom`: `Sphere.Refl.eq`, `evPt-subst`, `pointed-up`, `evPt-nat`
classes	`HomotopyClasses`: `STr`, `HC`, `compC`, `comp-assoc`, `pre-equiv`
π_3 on classes	`HomotopyGroups`: `pi3`, `pi3-map`, `pi3C`, `pi3C-comp`, `GIso`
Theorem 6	`SphereUniversal`: `Interfaces`, `Assemble.tu`, `ut`, `iso-S2-P`
loop isomorphisms	`LoopIso`: `loopPE`, `LoopEquivIso`, `ProjIso`, `LoopsSetIso`
Proposition 7	`PostnikovRealization.Triple`: `trunc5`, `conn3`, `s1`–`s5`, `pi3-iso`
(H3), (H4) as types	`KZ2Cohomology`: `Phi`, `H4`, `H3`
(H3), (H4) proved	`H3Proof.h3`; `CupSquare.kappa`, `H4Proof.h4`
Corollary	`Pi3Sphere.Conditional.Final.pi3-S2`

8. Audit

Assumptions. The `Theory` parameters of `sag/em` (univalence, images, surjectivity; function extensionality derived). The sphere axiom as a segment parameter. No other axiom; the checker was not modified; no choice principle.

Checking against the text (`seg/WRITING.md`, uniform checker `headlinear`, `-opened` and `-opens pi3.opens`). Every declaration checks in under 0.01 s. `c-refl`, `conn3`, `pi3-iso`, `iso-S2-P` open nothing beyond their text. The extras that remain: (i) `Refl.eq`, `evPt-nat`: the base point of a looping ΩF refl computes to refl (`Maps.D`, `Kf`, `idJR`), a definitional fact used in types; (ii) `tu`, `ut`, `comp-assoc`: the carrier $[X, Y]_*$ is opened to the quotient `Quot.Q` and its `image`; (iii) the π_1 lemmas open `PiOne.grp`/`QGrp.G` to `Quot.Q`, because the library states `Induced.map-cls` with `grp.G` while `Descend` works on `Q.Q`. (ii) and (iii) come from the interfaces of the shared library (`Quotient`, `QGrp`, `PiOne`), which were not edited. Aliases of named notions were removed (`Assemble` uses `open TR`, `open RP`; `Triple` writes `KA.K two`); `GIso` is a record with accessors.

Not done. π_2 as a group; $\mathrm{Hom}(\mathbb{Z}, B) \cong B$; the checks of the handoff on triples $(K(A,2) = (A,0,0)$, Fib0 $\simeq K(A,2) \times K(B,3)$, induced maps of squares on π_2, π_3, $\Phi_B(0) = 0$, $\Phi_{\mathbb{Z}}(\mathrm{id}) = [\kappa]$).